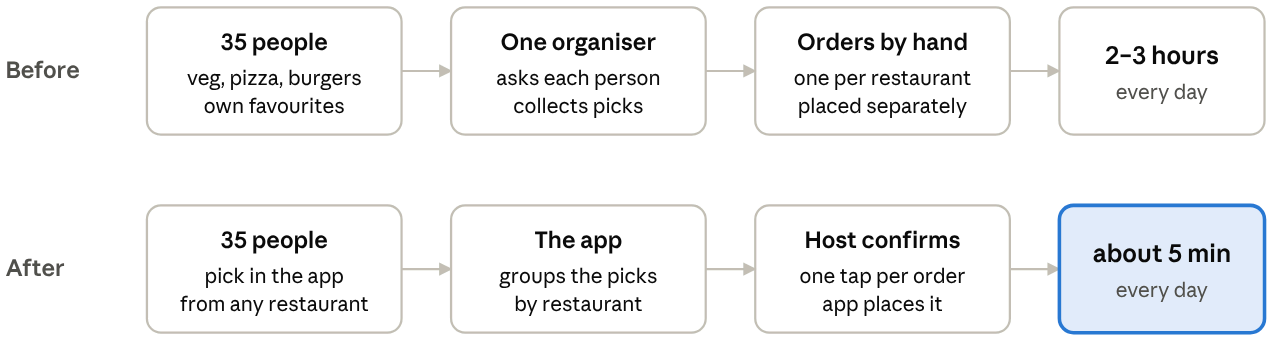
Multi-Restaurant Group Order App — Architecture Design

Nov 2, 2026 · @Fuhad Sanin

Problem

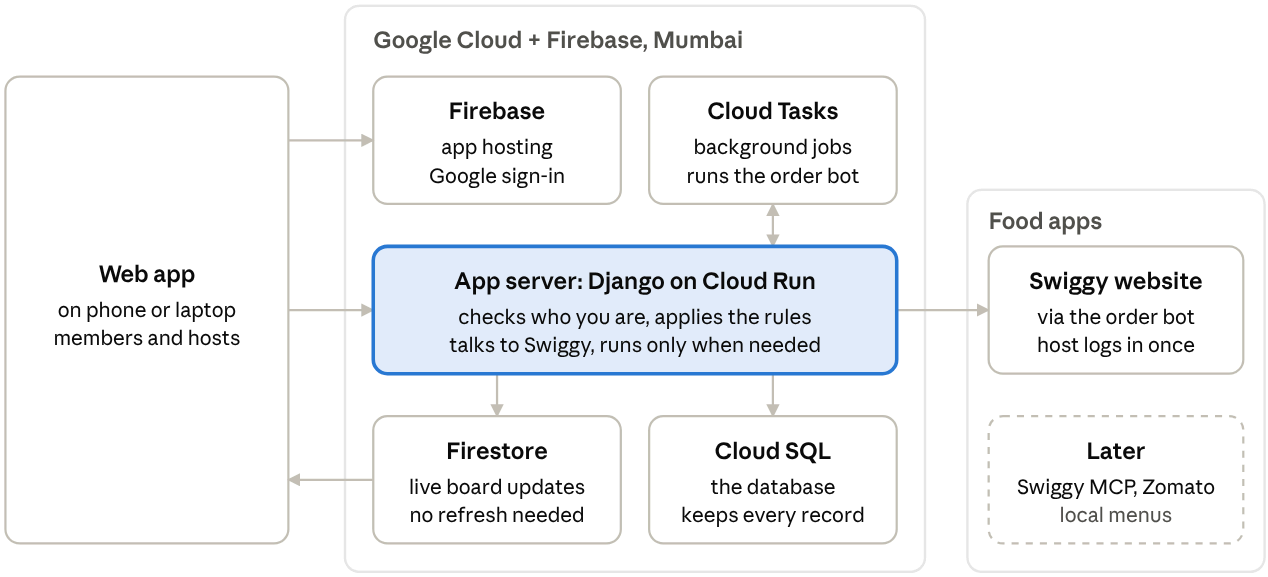
From 2–3 hours of collecting to about 5 minutes a day



Before and after · a 35-person team's daily lunch order

Swiggy's new multi-restaurant checkout (since 15 October 2026) is still one person's cart, capped at five restaurants. Collecting a whole team's picks remains the gap, so the plan is to host this for any team, then add local menus and other apps.

System architecture



System architecture · 11 components

Use cases: who does what

Everyone signs in with Google, then plays one of three roles.

Members pick food, the host runs the lunch, the admin sets up the company



Use case diagram · 4 actors, 10 use cases

Coming later: bill splitting, "same as last time", repeat lunches, company budgets, local menus and Zomato.

How a lunch order flows

Everyone picks, the host confirms, the app places one order

	Step	Who	What the app saves
1	**Host starts a lunch** chooses restaurants and a cutoff time, shares the link	Host	the lunch and its cutoff
2	**Everyone picks their food** each person chooses dishes from any listed restaurant	Everyone	each person's dishes
3	**Picks lock at the cutoff** the app puts everyone's dishes into one Swiggy cart	App	one cart, up to 5 restaurants
4	**Host checks and confirms** checks the cart and total, taps confirm once	Host	the host's OK
5	**The app places the order** one checkout on Swiggy, placed exactly once	App	order number and status
6	**Food arrives** everyone sees the status; the host hands out the food	Swiggy	delivery status

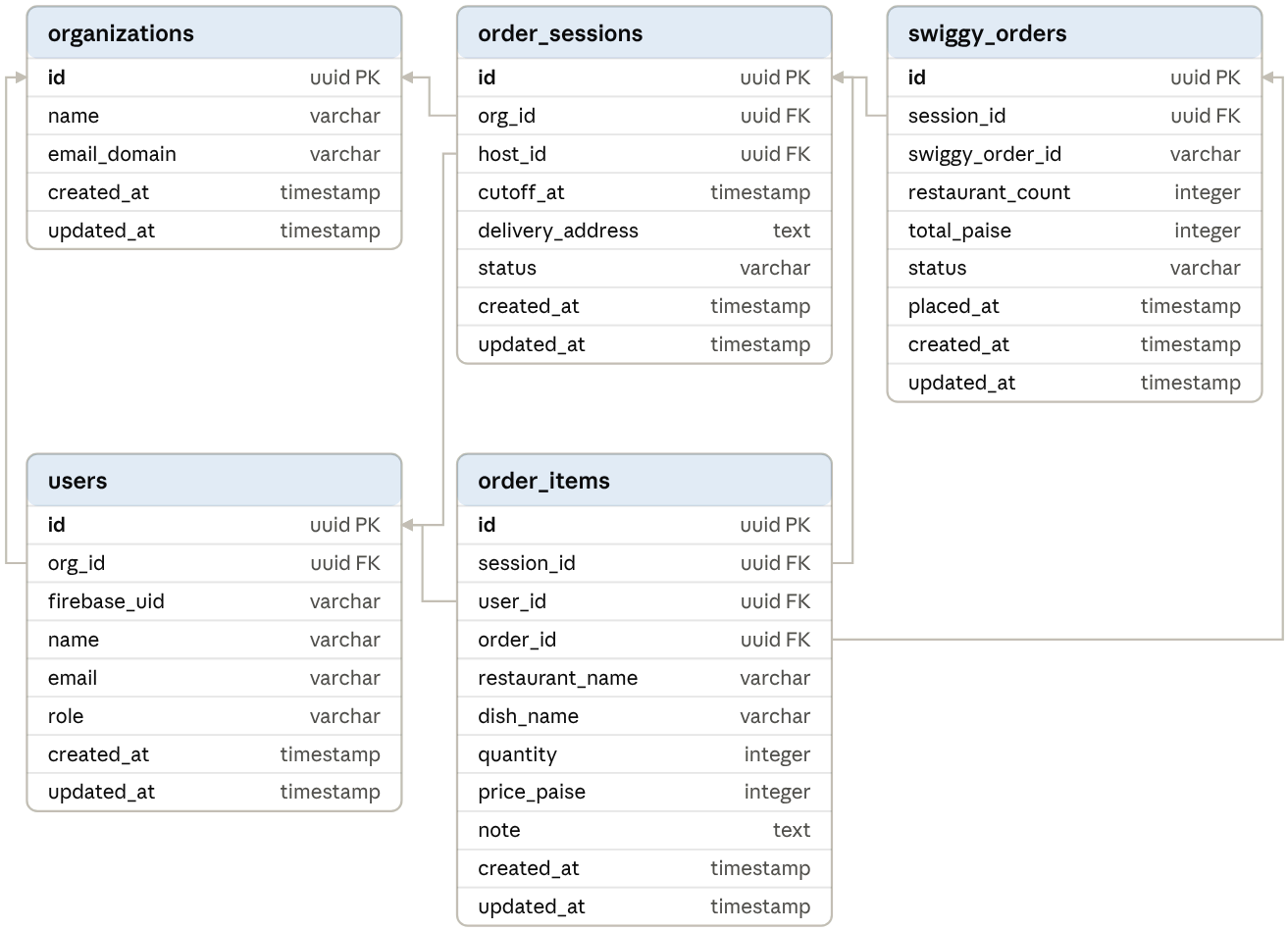
Lunch order flow · 6 steps, who does each, what the app saves

Today the order bot adds everyone's dishes to one Swiggy cart and places a single order, using Swiggy's multi-restaurant checkout (up to 5 restaurants per order; a lunch with more restaurants needs a second order). Once Swiggy approves MCP access, ordering moves to the MCP, which today takes one restaurant per cart and up to ₹1,000 per order.

Database schema

Five tables: people pick dishes in a lunch, placed as one Swiggy order

Every table has id, created_at and updated_at; FK lines show which id each column points to



Database schema · 5 tables with types and keys

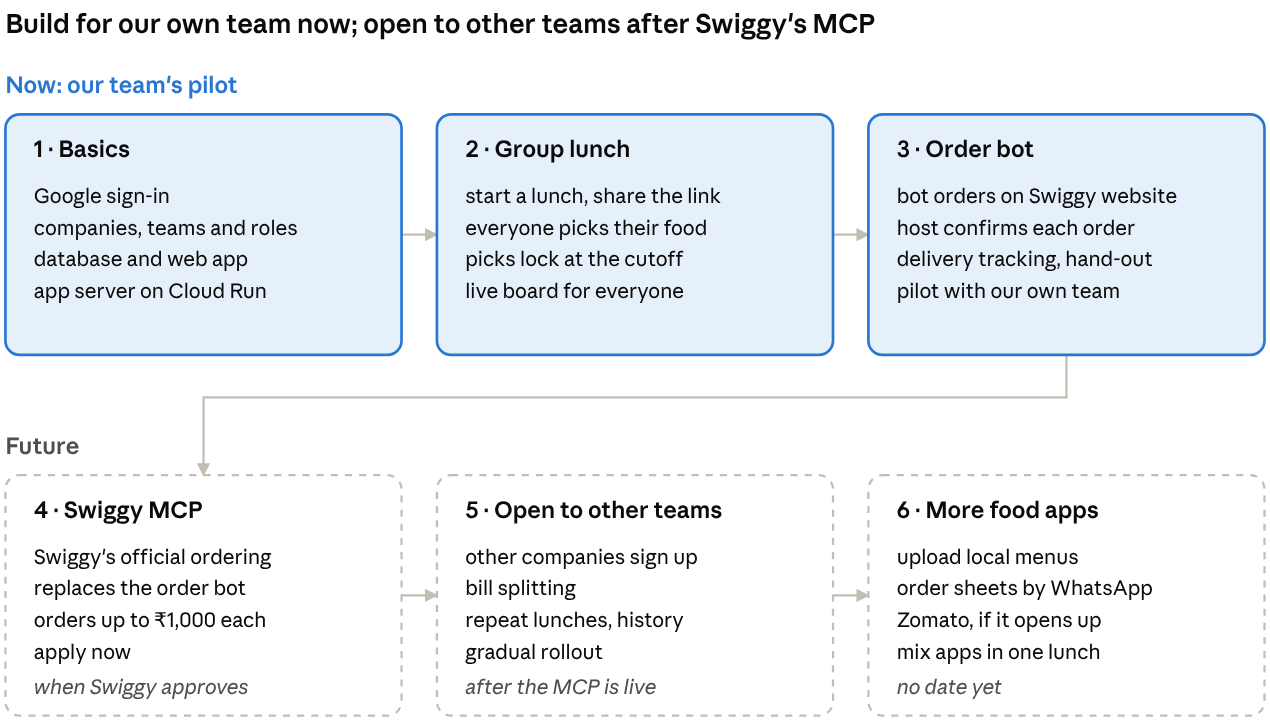
Security and reliability

Concern	How we handle it
Who is signing in	Google sign-in through Firebase; every request is checked
Company privacy	Each company sees only its own data, enforced in the database too
Swiggy logins	Stored encrypted; never sent to the browser

Concern	How we handle it
Ordering twice by mistake	Each order gets a one-time key, and the app checks Swiggy before any retry
Spotting problems	Error alerts, a log of every Swiggy call, and an alert when an order fails

Future scope and risks

Roadmap: now and later



Roadmap · 3 steps now, 3 in the future

Apply for Swiggy MCP access now: the review can take 4+ weeks, and until then the order bot is only for our own team.

Risks and what we do

Risk	What we do
Swiggy's rules don't allow bots, so the host's Swiggy account could be blocked	Use the bot only for our own team, a few orders a day; switch to the MCP once approved
Swiggy changes its website and the bot stops working	We get an alert; the host finishes the order in the Swiggy app
Swiggy's MCP approval can take 4+ weeks	Apply now, and say clearly that one host orders for a whole team
With the MCP, each order is capped at ₹1,000, so big orders split and pay more delivery fees	Ask Swiggy for a higher limit
Orders can't be cancelled from our app	The host cancels through Swiggy customer care
Zomato doesn't let outside apps order yet	Talk to Zomato early

At a glance: how it's built

Area	Part	What it does	Built with
Front end	Web app	Where people pick food and hosts run the lunch	React, hosted on Firebase
Front end	Sign-in	Log in with a Google account	Firebase Auth
Back end	App server	Checks who you are, applies the rules, starts each order	Django on Google Cloud Run
Back end	Database	Keeps companies, people, lunches and orders	PostgreSQL on Google Cloud SQL
Back end	Live board	Shows everyone's picks as they change	Firebase Firestore

Area	Part	What it does	Built with
Automation	Order bot	Opens Swiggy in a hidden browser, adds everyone's dishes to one cart and places the order on the host's account	Playwright with headless Chromium, on Cloud Run
Automation	Job queue	Starts the bot once the host confirms, then checks delivery	Google Cloud Tasks
Automation	Saved Swiggy login	Host logs in once with phone and OTP; the bot reuses that login	Encrypted, in Google Secret Manager
Automation	Double-order check	Before any retry, the bot looks at the host's Swiggy orders	Playwright
Alerts	Notifications	Tells people when to pick and when food arrives	Firebase Cloud Messaging, email
Hosting	Region	Runs in Mumbai, close to the people using it	Google Cloud, asia-south1

Sources

- Swiggy MCP: order food end-to-end
- Swiggy MCP: authenticate
- Swiggy MCP: ship to production
- Swiggy MCP: access and onboarding
- Swiggy MCP: changelog
- Swiggy multi-restaurant ordering (NewsBytes)
- Zomato MCP README
- Swiggy Builders Club
- Swiggy MCP: data and compliance
- Swiggy terms and conditions
- Google Cloud Run pricing

- Cloud SQL pricing
- Firebase pricing
- Supabase pricing
- Google Cloud free trial